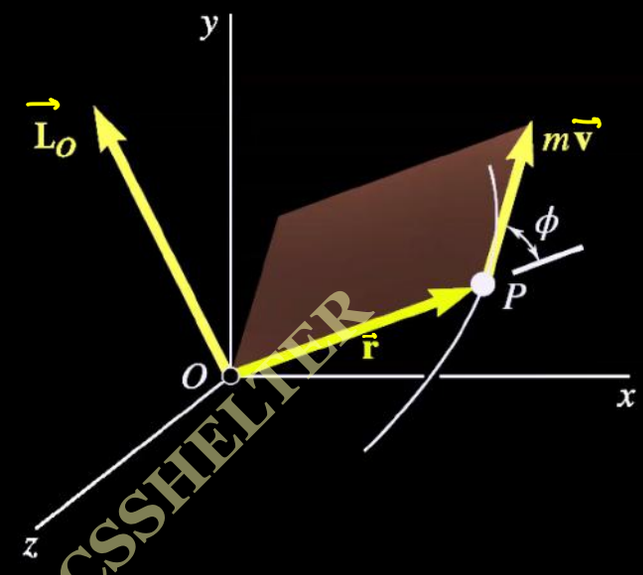
Rigid Body: A body in which distance between any two particles always remains fixed is called a rigid body.

Angular Momentum Of A Particle:



Let a particle P of mass m be moving with a velocity $\vec{v}$ in space. Further let $\vec{r}$ be its position vector wrt some origin O, then we define its angular momentum about point O,

$$\boxed{\vec{L_o} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v}} \quad (1)$$

when $\vec{p}$ is the linear momentum of the particle i.e $\vec{p} = m\vec{v}$.

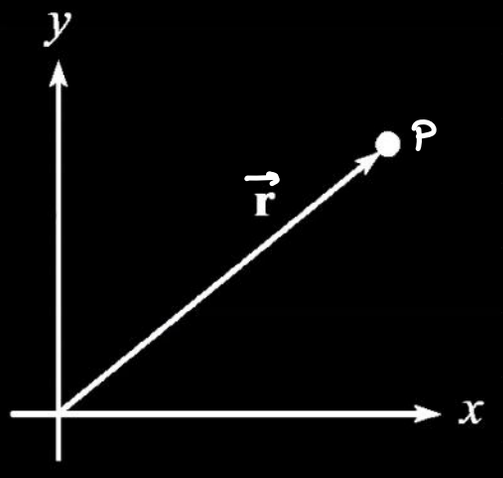
Thus:

$$\boxed{L_o = rmv\sin\phi = r_\perp mv = mrv_\perp}$$

when $r_\perp$ is component of $\vec{r}$ $\perp$ to $\vec{v}$ and $v_\perp$ is component of $\vec{v}$ $\perp$ to $\vec{r}$.

Torque As Time Derivative Of Angular Momentum:

Consider a point mass at position $\vec{r}$ relative to a fixed origin. The time derivative of angular momentum $\vec{L} = \vec{r} \times \vec{p}$ is:



$$\frac{d\vec{L}}{dt} = \frac{d}{dt}(\vec{r} \times \vec{p}) = \frac{d\vec{r}}{dt} \times \vec{p} + \vec{r} \times d\vec{p}/dt$$

$$= \vec{v} \times (m\vec{v}) + \vec{r} \times \vec{F}$$

$$= 0 + \vec{r} \times \vec{F}$$ where $\vec{F}$ is the net force acting on the particle. Now we define torque on a particle as $\boxed{\vec{\tau} = \vec{r} \times \vec{F}}$ then our relation between

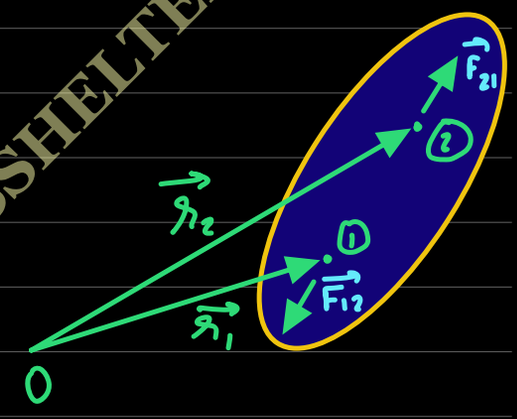
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torque and angular momentum becomes:

$\boxed{\vec{\tau}_O = d\vec{L}_O/dt}$ or in simple words, net torque of all the forces acting on a particle about some fixed point O is equal to rate of change of angular momentum about the same point O. (In case O is a point fixed in some non-inertial frame, we also need to consider torque of pseudo forces).

NET TORQUE OF INTERNAL FORCES:

Let ① and ② be two particles in a system of particles having position vectors $\vec{r}_1$ and $\vec{r}_2$ wrt some origin. Further let $\vec{F}_{12}$ be the force on particle 1 due to particle 2 and let $\vec{F}_{21}$ be force on particle 2 due to particle 1 then $\vec{F}_{12} = -\vec{F}_{21}$ and both the forces act along the line joining the particles according to Newton's third law. Now the sum of the torque of the action reaction pair about O becomes:



$$\vec{\tau}_p = \vec{r}_2 \times \vec{F}_{21} + \vec{r}_1 \times \vec{F}_{12}$$

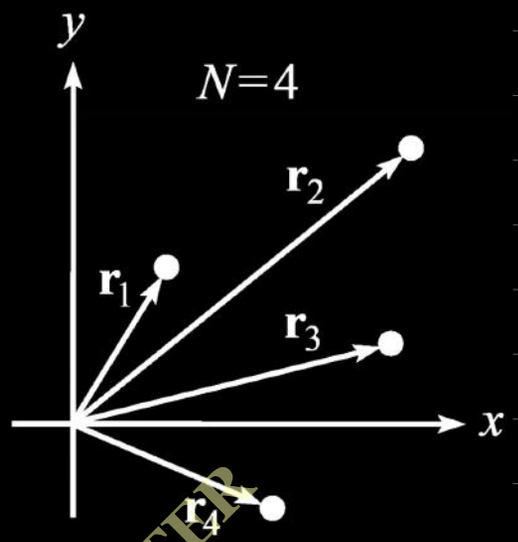
$$\Rightarrow \vec{\tau}_p = \vec{r}_2 \times (-\vec{F}_{12}) + \vec{r}_1 \times \vec{F}_{12} = (\vec{r}_1 - \vec{r}_2) \times \vec{F}_{12}$$

$$\Rightarrow \vec{\tau}_p = \vec{r}_{12} \times \vec{F}_{12}$$ but $\vec{r}_{12}$ and $\vec{F}_{12}$ are collinear by Newton's third law. Thus the cross product must be zero. Similarly summation of torques of all internal force action reaction pairs is zero, therefore if we have a system of particles then net torque on the system of particles is same as net torque of external forces on the system (∵ sum of internal torques is zero).

→ Summation of torques of internal forces in any system of particles is zero.

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Angular Momentum And Torque On A System Of Particles:



Angular momentum for a system of particles is simply the summation of angular momenta of its individual particles. Thus:

$\vec{L} = \sum \vec{r}_i \times \vec{p}_i$, now differentiating this, we get:

$$\frac{d\vec{L}}{dt} = \frac{d}{dt} \sum \vec{r}_i \times \vec{p}_i$$

$$= \sum \frac{d\vec{r}_i}{dt} \times \vec{p}_i + \vec{r}_i \times \frac{d\vec{p}_i}{dt}$$

$$= \sum \vec{v}_i \times (m_i \vec{v}_i) + \sum \vec{r}_i \times (\vec{F}_i^{ext} + \vec{F}_i^{int}).$$ (int: Internal ext: External).

But $\sum \vec{r}_i \times \vec{F}_i^{int} = 0$ (As shown in last section) thus, for a system of particles:

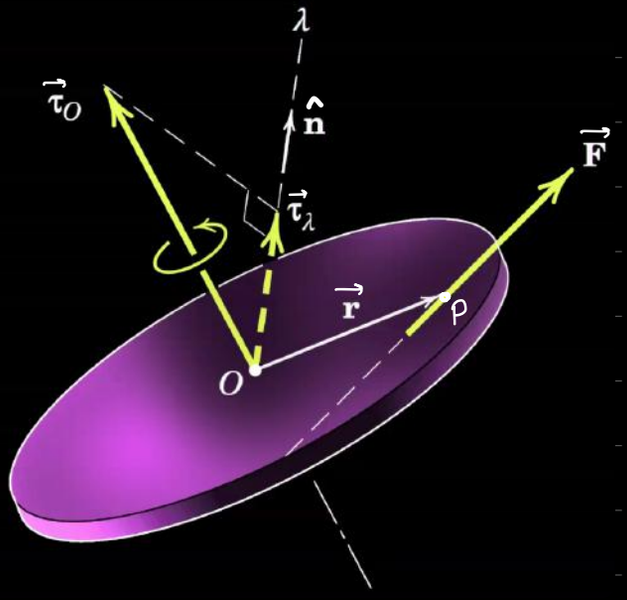
$$\boxed{\vec{\tau}_{ext} = \frac{d\vec{L}}{dt}}$$

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Torque About An Axis:

Let λ be some axis along a unit vector $\hat{n}$ and let O be some point on the axis λ. Then we define torque of force $\vec{F}$ about axis λ as

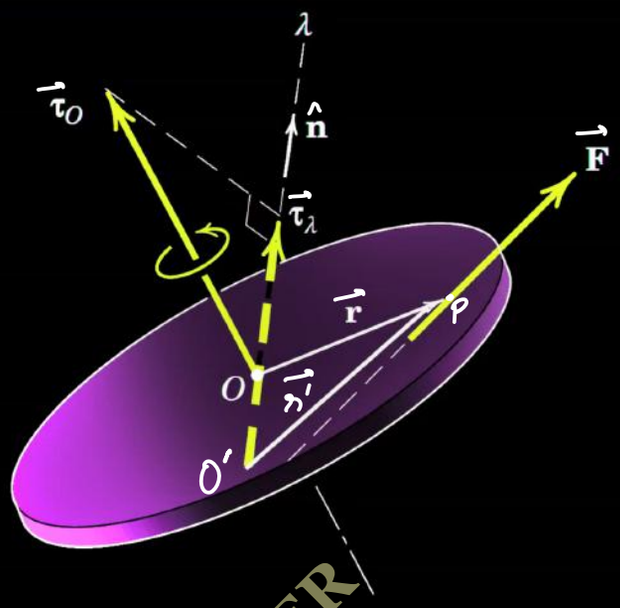
$$\boxed{\vec{\tau}_\lambda = [(\vec{r} \times \vec{F}) \cdot \hat{n}]\,\hat{n}}$$ (1) thus



vector component of torque about point O along the axis is the torque about the axis i.e:

$$\boxed{\vec{\tau}_\lambda = [\vec{\tau}_O \cdot \hat{n}]\,\hat{n}}$$ (2)

→ One might think that this definition of torque about an axis is ambiguous because there are infinite number of points on an axis and one could choose any of these points for defining torque about the axis, however we will show that we will get the same result irrespective of choice of point.



Let us choose another point O' on λ for defining torque about the axis then:

$$\vec{\tau}'_\lambda = [(\vec{r}' \times \vec{F}) \cdot \hat{n}]\hat{n} \quad (3)$$

Now subtracting (3) from (1) and using distributive properties of dot and cross products,

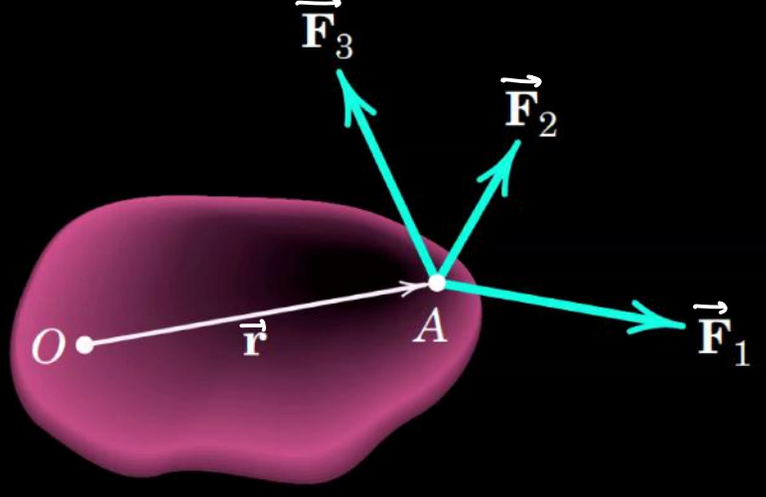
$$\vec{\tau}'_\lambda - \vec{\tau}_\lambda = [(\vec{r}' - \vec{r}) \times \vec{F} \cdot \hat{n}]\hat{n} = [(\vec{O'O} \times \vec{F}) \cdot \hat{n}]\hat{n}$$

Now recognize the $\vec{O'O}$ is along the axis thus $\vec{O'O} \times \vec{F}$ is $\perp$ to $\hat{n}$ and hence $(\vec{O'O} \times \vec{F}) \cdot \hat{n} = 0$ thus: $\vec{\tau}'_\lambda - \vec{\tau}_\lambda = 0$.

→ It does not matter, which point on the axis we choose to define torque about the axis.

VARIGNON'S THEOREM:

If several forces are acting at a point then their net torque can be found by simply taking the torque of their resultant (Conversely, torque of a force is equal to sum of torques of its components).



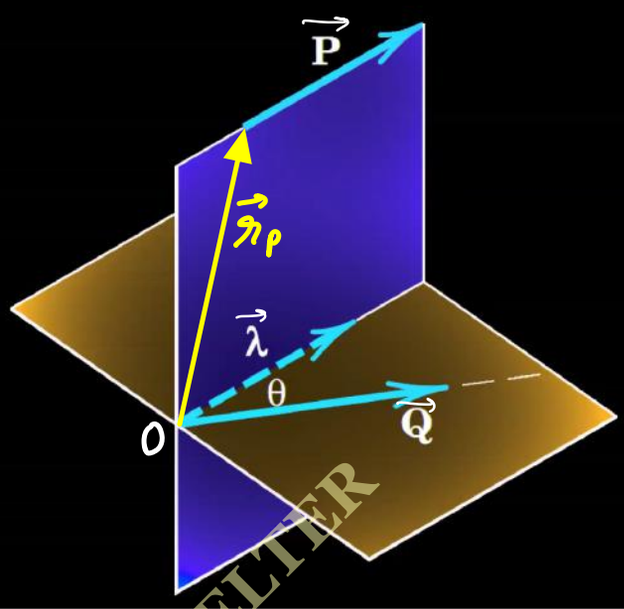
This can be seen readily as:

$$\vec{r} \times \vec{F}_1 + \vec{r} \times \vec{F}_2 + \vec{r} \times \vec{F}_3$$
$$= \vec{r} \times (\vec{F}_1 + \vec{F}_2 + \vec{F}_3) \quad \because \text{Distributive property of cross product.}$$

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Torques Of Intersecting And Parallel Forces About An Axis:

If a force is parallel to an axis or its line of action passes through the axis then it produces zero torque about the axis. For instance in adjacent figure, the force $\vec{Q}$ intersects with the axis λ. Now to find the torque we could choose point O as the intersection point itself. Thus $\vec{r}$ and $\vec{Q}$ are coincident and hence $\vec{r} \times \vec{Q}$ becomes zero. Similarly consider force $\vec{P}$ which is $\parallel$ to axis λ. Clearly from the figure we can see that $\vec{r}_p \times \vec{P}$ is perpendicular to $\vec{\lambda}$ ($\because \vec{P}$ is $\parallel$ to $\vec{\lambda}$). Thus $(\vec{r}_p \times \vec{P}) \cdot \hat{\lambda} = 0$, Thus:

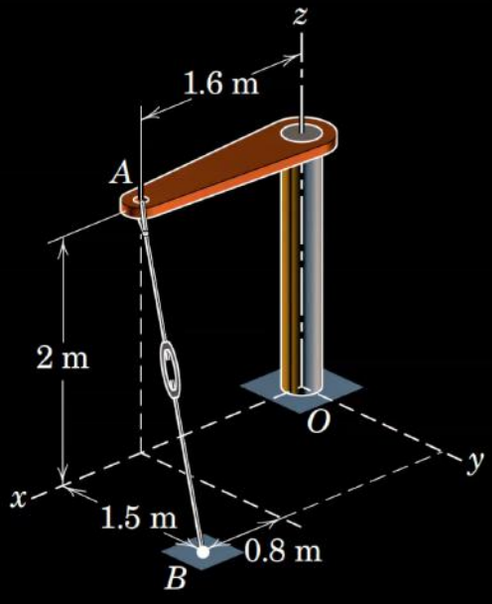


→ Intersecting and parallel forces (or components of forces) produce no torque about an axis.

→ Any Arbitrary force $\vec{F}$ acting in 3D space can be resolved in to 3 components wrt any arbitrary axis λ:

1. Parallel $\vec{F}_{\parallel}$
2. Intersecting Perpendicular
3. Skew Perpendicular

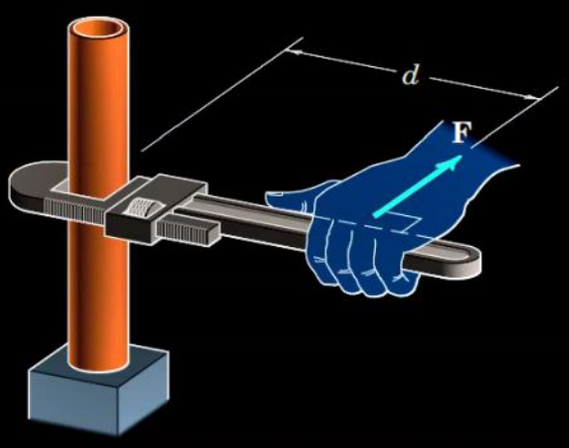
For instance suppose we want to find torque due to tension of the cable acting at point A about Z axis then x component of tension will be intersecting $\perp$, z component will be $\parallel$ and y component will be skew $\perp$. Thus we only need to find torque of the y component of the force.



→ If $r_{\perp}$ is the perpendicular distance of line of action of a skew $\perp$ force then its torque magnitude is simply $r_{\perp} F$.

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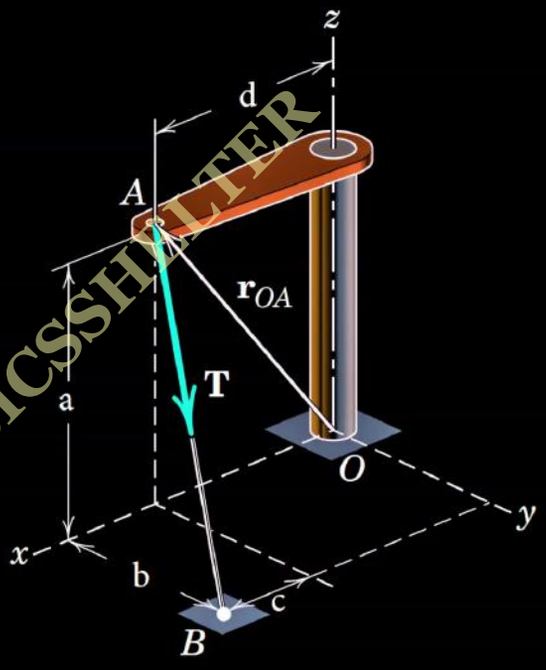
For instance, in adjacent figure, the torque is simply $\tau = Fd$. (The force is already shew $\perp$ so no need to resolve it in to components. Thus are no $\parallel$ and intersecting components here.



EXAMPLE: In the adjacent figure, point A is being pulled by a tension T as shown. Find the torque of this force about z axis.

Solution: If we want we could do $[(\vec{r}_{OA} \times \vec{T}) \cdot \hat{k}]\,\hat{k}$, but it will be a bit cumbersome. So to simplify our task we will simply take the torque of shew $\perp$ component, i.e y component of $\vec{T}$. Thus, $\vec{T}_y = \dfrac{Tb}{\sqrt{a^2+b^2+c^2}}\,\hat{j}$

thus required torque, $d\hat{i} \times \vec{T}_y = \dfrac{Tbd}{\sqrt{a^2+b^2+c^2}}\,\hat{k}$ Ans.

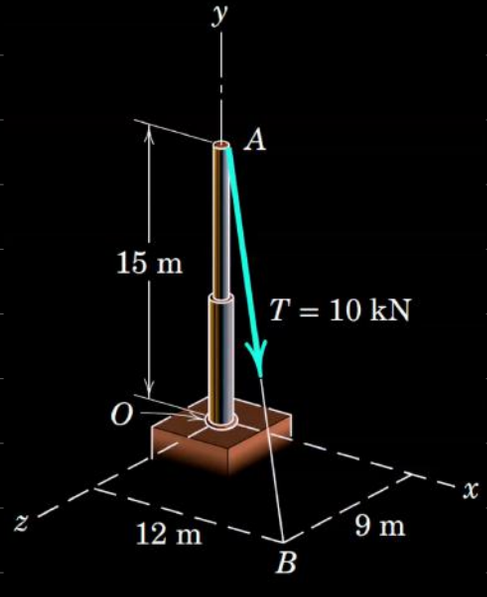


EXAMPLE: For the situation shown in the adjacent figure, find the torque of the tension T about z axis.

Solution: Here x component is the shew $\perp$ component. Thus the require torque:

$$15\hat{j} \times 10 \times \frac{12}{\sqrt{12^2+9^2+15^2}} \times 10^3\,\hat{i}$$

$$= 84.9\ \text{kNm}\ (-\hat{k})\ \text{Ans.}$$



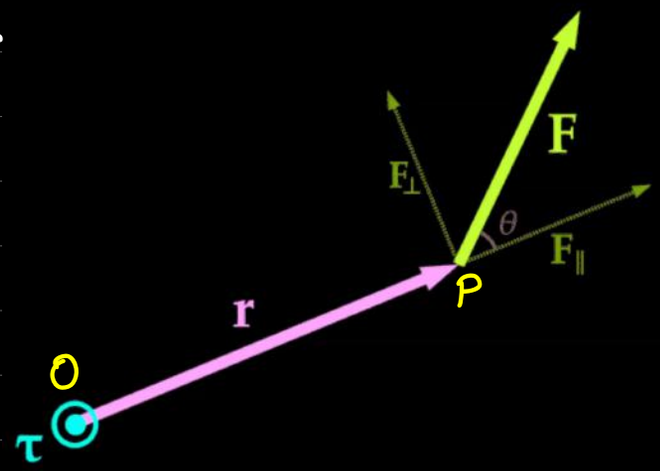
(Sometimes Joule is also taken as a unit of torque).

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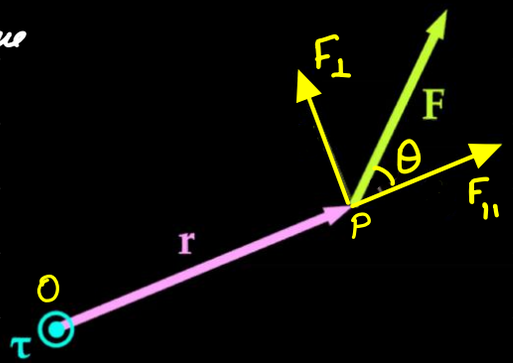
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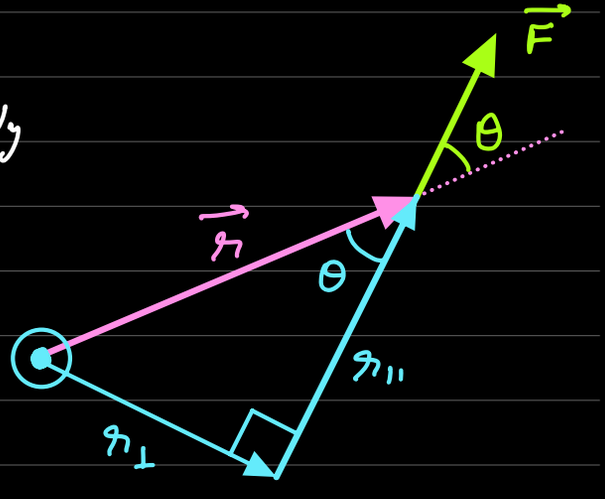
Torque As A Signed Scalar:

Even though torque is a vector, for problems involving forces only in two dimensions, we may treat it as a signed scalar. In the plane of the page, torque of any force about any point may be considered as either clockwise or anti-clockwise. (We can see this by imagining a stick pivoted at point about which we are finding the torque with other end at point of application of force. Now visualise the effect of the force on this pivoted stick). For instance, in the figure shown above, the torque of $\vec{F}$ about O is anticlockwise as can be seen by imagining the stick OP (pink). If clockwise torques are assumed to be positive then, anti-clockwise torques must be considered negative and vice versa.



Magnitude Of Torque: Recall that the torque $\vec{\tau} = \vec{r} \times \vec{F}$ thus using definition of cross product $\tau = rF\sin\theta$, now if we resolve F in to components $\parallel$ and $\perp$ to r we recognize that $F\sin\theta$ is nothing but component of $\vec{F}$ $\perp$ to $\vec{r}$. Thus we can also write $\boxed{\tau = rF_\perp}$ or simply "distance times perpendicular force." Alternatively we could also resolve $\vec{r}$ in to components $\parallel$ and $\perp$ to F and we recognize that $r_\perp = r\sin\theta$, thus $\tau = rF\sin\theta = Fr\sin\theta = Fr_\perp$ thus another way of writing torque is $\boxed{\tau = Fr_\perp}$ or simply,



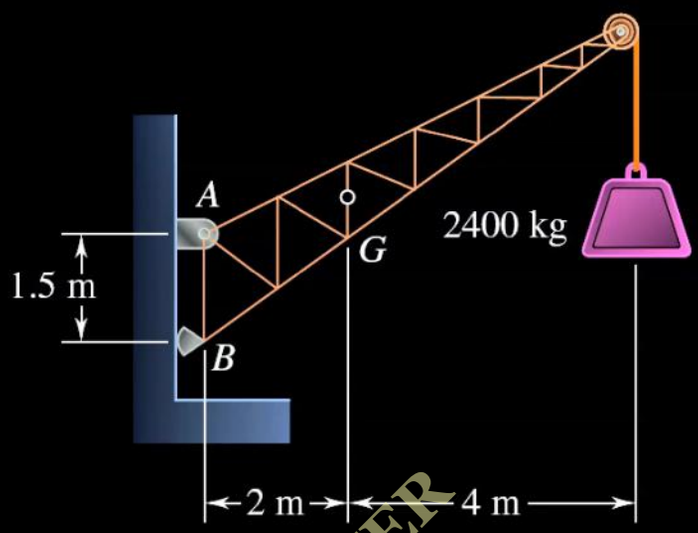


Torque is Force times perpendicular distance.

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Equilibrium Of A Still Rigid Body:

Recall that we showed for any system of particles that $\vec{\tau} = d\vec{L}/dt$. Now if a rigid body is still (both velocity and acceleration is zero for every particle) then its angular momentum is constantly zero about every point of the frame from which it appears still. Thus rate of change of its angular momentum is zero and hence torque about every point of the frame from which the rigid body appears still must be zero. Thus for equilibrium of a still body:



$$\boxed{\sum \vec{\tau}_{ext} = 0} \quad \text{also,} \quad \boxed{\sum \vec{F}_{ext} = 0}$$

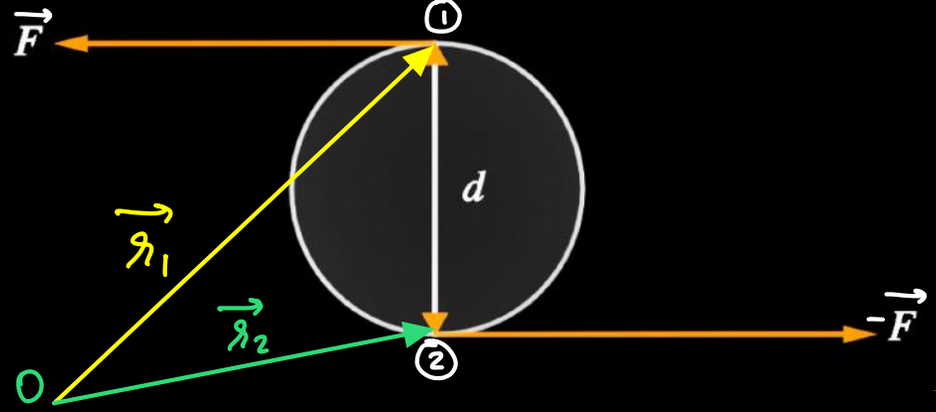
Ext: External.

→ In case the frame from which rigid body appears to be still, is a non-inertial frame, then we also need to consider the torque of pseudo forces.

→ If in some frame rigid body appears to be moving in pure translation with uniform velocity then in such a frame also, net torque must be zero because angular momentum remains constant (although non-zero).

Couple:

Two equal and opposite forces acting on a rigid body form a couple. The net torque of a couple is independent of point about which torque is being considered.



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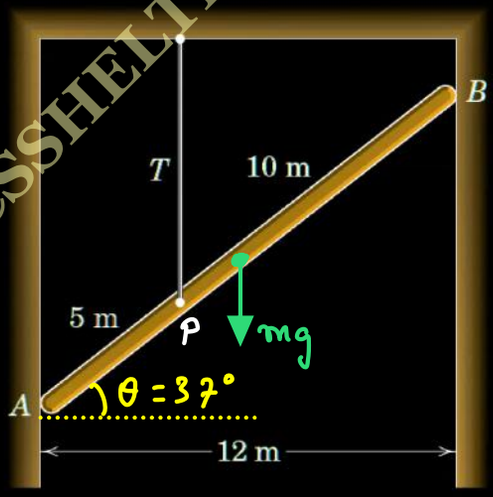
This can be easily seen as follows. Consider the forces $\vec{F}$ and $-\vec{F}$ acting on a disc having position vectors $\vec{r}_1$ and $\vec{r}_2$ wrt some arbitrary point O. Then, net torque about O:

$$\vec{M} = \vec{r}_1 \times \vec{F} + \vec{r}_2 \times (-\vec{F}) = (\vec{r}_1 - \vec{r}_2) \times \vec{F} = \vec{r}_{12} \times \vec{F}$$

now $\vec{r}_{12}$ is simply position vector of ① wrt ② which is independent of choice of point O. Thus net torque of a couple is same about every point QED.

→ Torque is also called moment, hence the symbol $\vec{M}$.

EXAMPLE: The uniform 15m pole has a mass of 150 kg and is supported against its smooth ends by vertical walls and tension T. Find the reactions at A and B.



Solution: The reactions at A and B form a couple. Now let the reactions at A and B be N each. Balancing torques about point P:

$$1500 \times (7.5 - 5)\cos 37^\circ = N \times 15 \sin 37^\circ \Rightarrow N = 100 \times 2.5 \times \frac{4}{3}$$

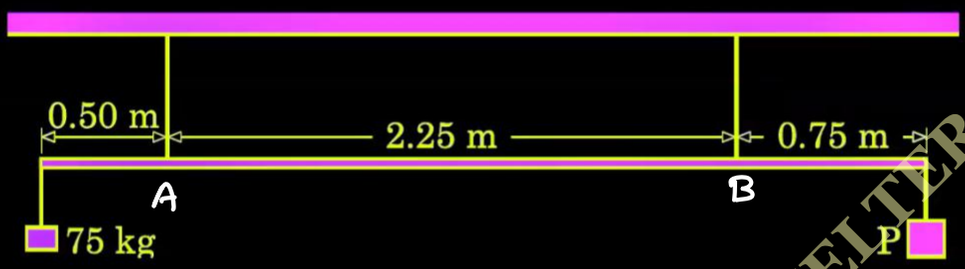
$$= 333.33 \text{ N} \quad \text{Ans.}$$

CONCURRENCY THEOREM: If a body is in equilibrium under action of three forces then the three forces must be concurrent (or parallel i.e. concurrence at ∞). This can be proven very easily. Suppose lines of actions of two forces intersect at some point, then torque of those two forces will be zero about that point. Now if third force does not pass through that point then its torque will be non zero thus producing a non-zero resultant torque.

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Example: Pathfinder: MCQs:

A light beam suspended with the help of two ropes supports two loads as shown in the figure on the next page. The load at the left end is of 75 kg and the load P at the right end is unknown. The maximum tension that either of the rope can bear is 1200 N. In an experiment, mass of the load P is gradually increased from zero. If the rod remains horizontal in equilibrium, which of the following conditions must be satisfied?



(a) Minimum mass of the load P can be 12.5 kg.
(b) Maximum mass of the load P can be 102.5 kg.
(c) When mass of the load P is minimum, tension in one of the ropes vanishes.
(d) When mass of the load P is maximum, tension in one of the ropes vanishes.

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Solution: As the mass of P is gradually increased, there will be a point when the right rope is just taut (i.e $T=0$). At this point tension in the left rope is $T_L = 750N + m_p \times g$. Now balancing torques about P:

$$750\,(0.5 + 2.25 + 0.75) = (750 + m_p g) \times (2.25 + 0.75)$$

$\Rightarrow m_p = 12.5$ kg, thus at this point $T_L = 750 + 125 = 875\,N < 1200N$ thus, option A is correct, also C is correct.

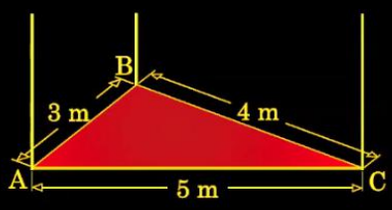
Now consider torque about contact point of right rope with beam. As P is made heavier, its clockwise torque increases therefore tension in the left string must decrease and $\therefore$ tension in the right string must increase (due to force balance). For beam to remain Horizontal, tension in the left string should be greater than zero (zero in limiting case). Now balancing torques about point B in limiting case with $T_L = 0$ (L: Left string): $m_p g \times 0.75 = 750 \times 2.75$

$\Rightarrow m_p = 275$ kg, But this makes $T_L = (275 + 750) \times 10$ i.e 3500 N which is more than the maximum allowable of 1200 N.

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Now let us consider the other possibility of $T_R = 1200\,N$ exactly
Now balancing torques about point A: $750 \times 0.5 + 1200 \times 2.25 = m_p g \times 3$
$\Rightarrow m_p = 102.5\,Kg$. Thus maximum value of P can be $102.5\,Kg$.
Thus B is correct. Correct options: A, B & C.

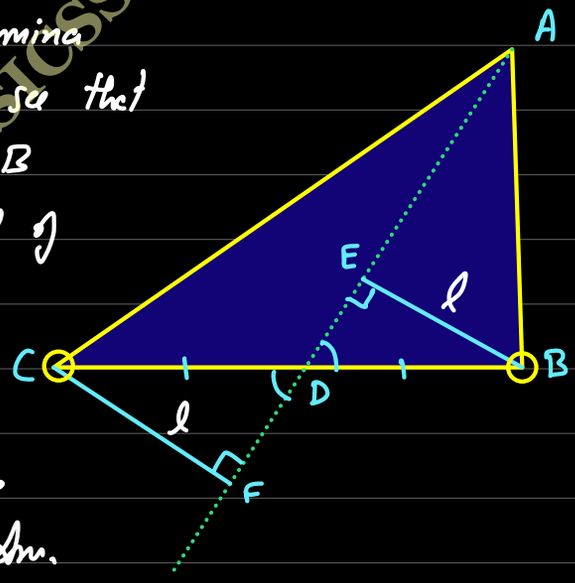
Example: Pathfinder MCQ 6:



A uniform right-angled triangular lamina is suspended from the ceiling with the help of three light strings. The plane of the lamina is horizontal and the strings are vertical as shown in the figure. In which of the following options, tensile forces T_A, T_B and T_C developed in the strings connected to the vertices A, B and C respectively of the lamina are arranged in correct order of their magnitudes?

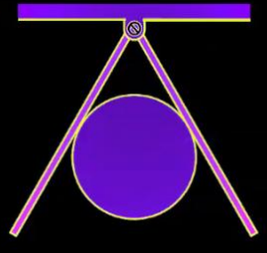
(a) $T_A > T_B > T_C$ (b) $T_B > T_A > T_C$
(c) $T_C > T_A > T_B$ (d) $T_A = T_B = T_C$

Solution: Let AD be the median of the lamina
By similar triangles CFD and BEF we can see that perpendicular distances of the vertices C and B from the median are equal. Also centroid of the lamina (CM) lies on the median.
Thus balancing torques about the median,
$T_C \times l = T_B \times l$ (∵ Both are skew 1)
$\Rightarrow T_B = T_C$ similarly using other medians we conclude that $T_A = T_B = T_C$ i.e option (d) Ans.



Example: Pathfinder: Build: 19:

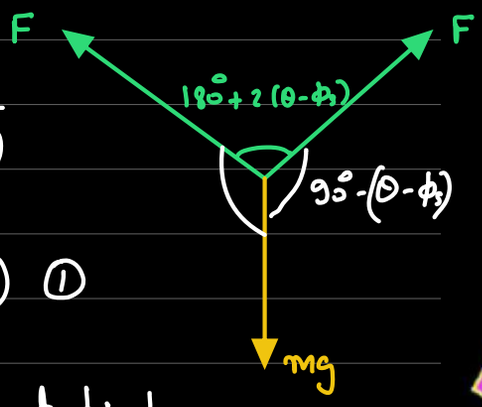
Upper edges of two identical uniform planks each of mass $M = 87$ kg and length $l = 1.0$ m are hinged to a horizontal axle affixed to the ceiling. A cylinder of mass $m = 20$ kg and radius $r = 0.30$ m is inserted horizontally between the planks as shown. The cylinder touches the planks at a distance $\eta = 0.75$ times the length of the plank from the hinge. Find range of coefficient of friction between the planks and the cylinder to keep the system in equilibrium.



Solution: The cylinder is in equilibrium under action of three forces i.e its weight and the two contact forces from the planks.
The cylinder has a downward slipping tendency, therefore the friction forces act upwards (along the planks).

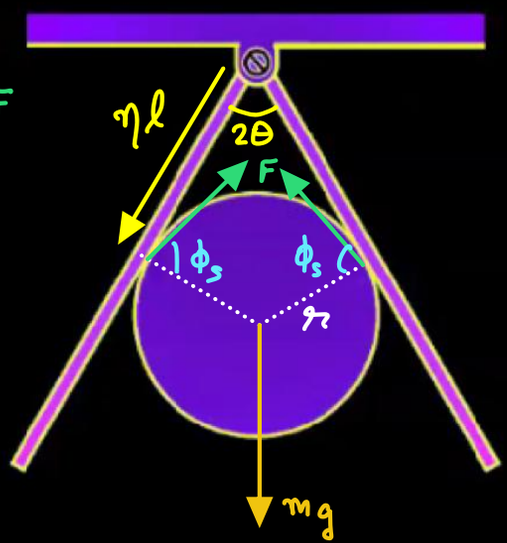
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Now applying Lami's theorem to the cylinder, (limiting equilibrium):



$$\frac{mg}{\sin 2(\phi_s - \theta)} = \frac{F}{\cos(\theta - \phi_s)}$$

$$\Rightarrow F = mg / 2\sin(\phi_s - \theta) \quad (1)$$



Also Balancing torques about pivot for the left plank:

$$F\cos\phi_s \times \eta l = Mg\frac{l}{2}\sin\theta \Rightarrow F = \frac{Mg\sin\theta}{2\eta\cos\phi_s} \quad (2)$$

Also $\tan\theta = r/\eta l$ From (1) & (2):

$$\frac{m}{\sin(\phi_s - \theta)} = \frac{M\sin\theta}{\eta\cos\phi_s} \Rightarrow \frac{\sin(\phi_s - \theta)}{\cos\phi_s} = \frac{\eta m}{M\sin\theta}$$

$$\Rightarrow \frac{\sin\phi_s\cos\theta - \cos\phi_s\sin\theta}{\cos\phi_s} = \eta\frac{m}{M\sin\theta}$$

$$\Rightarrow \mu\cos\theta - \sin\theta = \frac{\eta m}{M\sin\theta} \Rightarrow \mu = \tan\theta + \frac{\eta m}{M\sin\theta\cos\theta}$$

$\sin\theta = \dfrac{r}{\sqrt{r^2 + \eta^2 l^2}}$, $\cos\theta = \dfrac{\eta l}{\sqrt{r^2 + \eta^2 l^2}}$, thus,

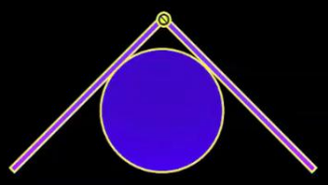
$$\mu = \frac{r}{\eta l} + \frac{m}{M}\left[\frac{r^2 + \eta^2 l^2}{rl}\right] = 0.9$$

[This is considering limiting equilibrium. Thus $\mu \geq 0.9$ Ans].

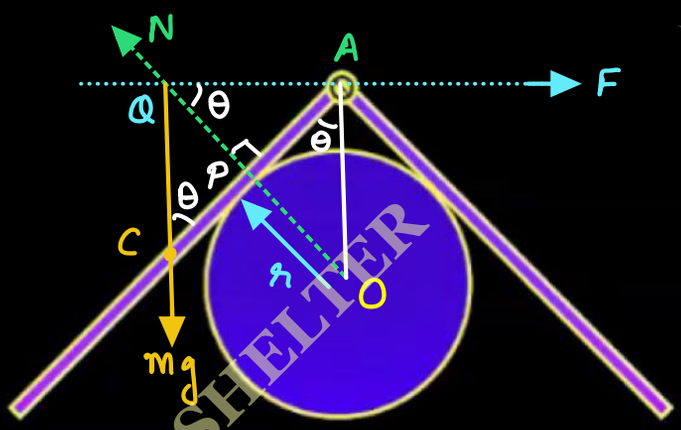
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EXAMPLE: PATHFINDER: BUILD20:

A structure consisting of two identical square plates of side $l = 40$ cm hinged to a common axle at one of their edges is placed on a fixed horizontal cylinder of radius $r = 10$ cm as shown in the figure. There is no friction between the cylinder and the plates. At what angle with vertical will the plates lean in equilibrium?



<u>Solution:</u> Due to left right symmetry, the hinge reaction must be horizontal. Also the plates are in equilibrium under action of three forces, therefore, the three forces must be concurrent. Now $AC = l/2$ ($\because$ (M), Also,



Now $AN = \frac{l}{2}\sin\theta = AO\cot\theta$

$= r\,\text{cosec}\,\theta\cot\theta$

Thus:

$$\frac{l}{2}\sin\theta = r\,\text{cosec}\,\theta\cot\theta \Rightarrow \frac{\sin^3\theta}{\cos\theta} = \frac{2r}{l} = \frac{1}{2}$$

$$\Rightarrow \quad 2\sin^3\theta - \cos\theta = 0 \qquad \text{Putting } \sin\theta = x:$$

$$2x^3 = \sqrt{1-x^2}$$

$$4x^6 + x^2 - 1 = 0 \qquad \text{put } x^2 = y$$

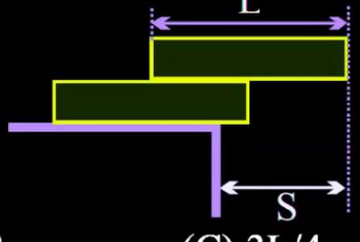
$$\Rightarrow \quad 4y^3 + y - 1 = 0$$

$$(2y-1)(2y^2 + y + 1) = 0 \quad \Rightarrow \quad y = 1/2 \Rightarrow \sin^2\theta = 1/2$$

$$\Rightarrow \theta = 45^\circ \text{ Ans.}$$

EXAMPLE:

Two identical bricks of length L are piled one on top of the other on a table as shown in the figure. The maximum distance S the top brick can overhang the table with the system still balanced is:

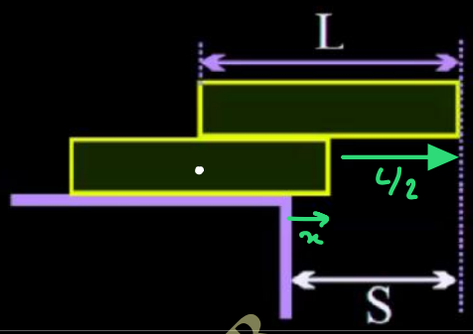


(A) L/2 (B) 2L/3 (C) 3L/4 (D) 7L/8

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Solution: To avoid toppling centre of mass of the overhanging portion at any level must be within edge of the previous block otherwise toppling will happen due to torque of gravity. Thus relative overhang of top block must be $\frac{L}{2}$.

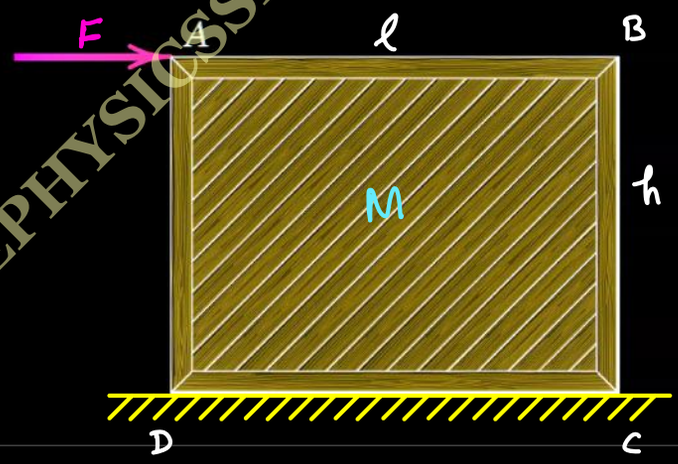
Now if relative overhang of lower block is x, Then making combine CM coordinate w.r.t table edge equal to zero (limiting case):



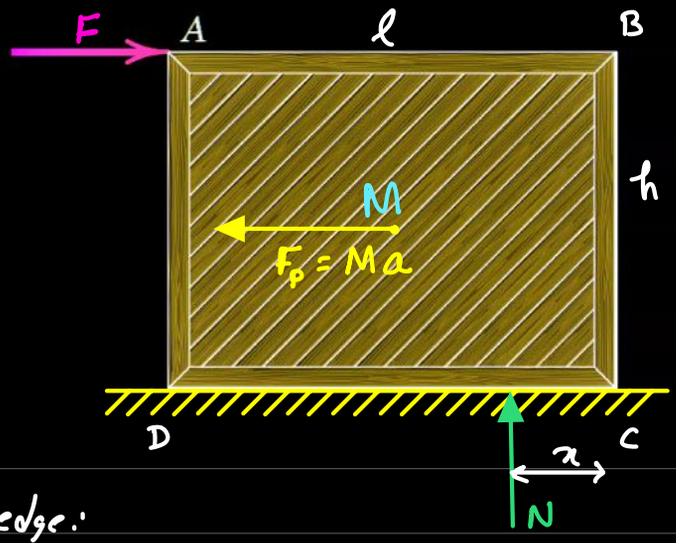
$$m(x) + m(x - L/2) = 0$$

$$\Rightarrow x = L/4 \Rightarrow \text{Total achievable overhang: } S = L/4 + L/2 = 3L/4 \quad \text{Ans (C).}$$

EXAMPLE: A box of mass M height h and width ℓ is being pushed by a force as shown in the figure. Where does the normal reaction act if the box is not toppling? For what maximum value of F does the box not topple.



Solution: Acceleration of the box is F/M. The box appears to be in equilibrium from its own frame. ∴ we must apply a pseudo force F_p at its CM where $F_p = Ma = \frac{MF}{M} = F$.



Now by vertical force balance, $N = Mg$. Let the normal reaction act at a distance x from the front edge. Balancing torques about front edge:

$$Fh + Nx = Mg\frac{\ell}{2} + Ma\frac{h}{2} \Rightarrow Fh + Mgx = Mg\frac{\ell}{2} + F\frac{h}{2}$$

$$\Rightarrow x = \frac{\ell}{2} - \frac{Fh}{2Mg} \quad \text{Ans.}$$

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Now at the verge of toppling, entire normal reaction must get shifted to C, thus $x = 0$

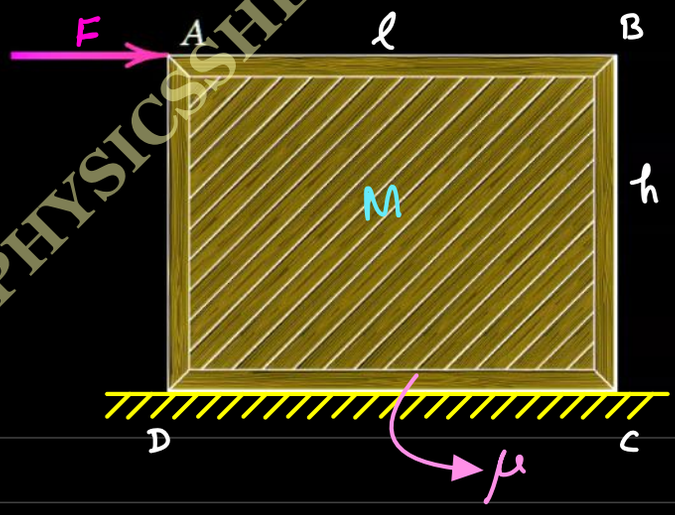
$\Rightarrow F = Mg\ell/h$

→ Caution: Remember to apply pseudo force before writing torque balance equations. After applying pseudo force, in the rest frame of the body we can balance torque about any point.

→ Since for a translationally accelerated body, pseudo force is applied at CM so it creates zero torque about CM. Thus if we wish to make torque equation about CM, we can still balance the torque of real forces (Provided there is only translational acceleration).

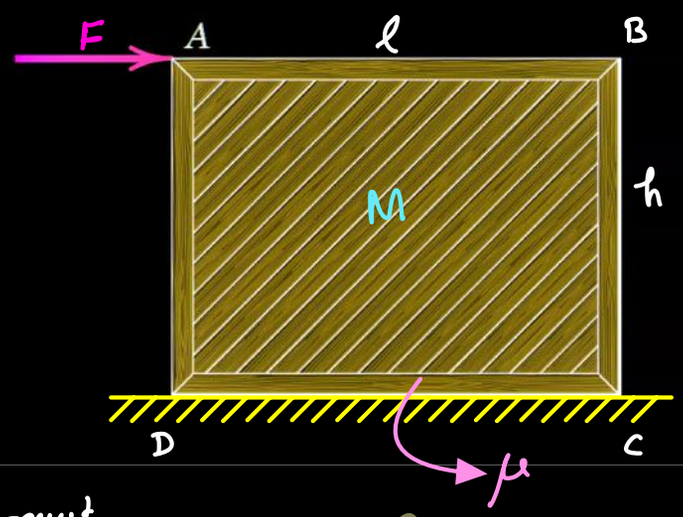
TOPPLING Vs SLIPPING:

This is a category of problems in which some parameter of a body in equilibrium is gradually varied until either slipping or toppling occurs. For instance in adjacent figure, we might apply a gradually increasing force and want to know whether the box will first topple or slip. (Note that surface is rough here). In such questions first we find the required value of the parameter (here Force) for slipping (assuming no toppling) and then we find required value of the parameter for toppling (assuming no slipping). Whichever value of the parameter is achieved first, the phenomenon corresponding to that assumption will occur first. This will be clear from the following example:



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EXAMPLE: In the shown set up, $\mu = 1/5$, $h/l = 3/5$ and a gradually increasing force is applied. Will the box begin to slip or topple first?



Solution: Assuming no toppling, force required for slipping is $F_s = \mu Mg = Mg/5$, Now for toppling, entire normal reaction must shift to the forward bottom edge. Assuming no slipping and balancing torques about edge through C at the verge of toppling:

$$Fh = Mg\,l/2 \Rightarrow F = Mg\frac{l}{2h} = Mg\frac{5}{6} = \frac{5}{6}Mg = F_T$$

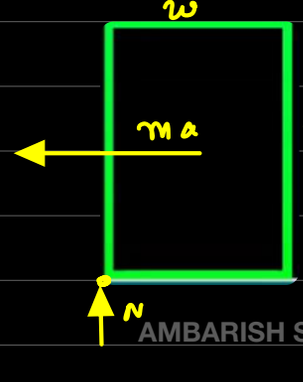
Thus we see that force required for toppling $F_T = \frac{5}{6}Mg$ which is more than force required for slipping $F_s = Mg/5$ therefore the block will first slip.

EXAMPLE: A flat bed truck carries an Almirah. The height to width ratio of Almirah is 3:2. The coefficient of friction on the bed of the truck is 0.5. If the truck moves with a variable acceleration, is the Almirah likely to topple or slip?



Solution: Assuming no toppling, the slipping happens when pseudo force exceeds limiting friction. Thus for slipping, $ma = \mu mg \Rightarrow a = \mu g$ (limiting case). $= 0.5g$

Now for toppling, torque of pseudo force must exceed that of gravity about the rear edge. Thus:



$$ma \times \frac{h}{2} = mg \times w/2 \Rightarrow a = \frac{w}{h}g = 2g/3$$

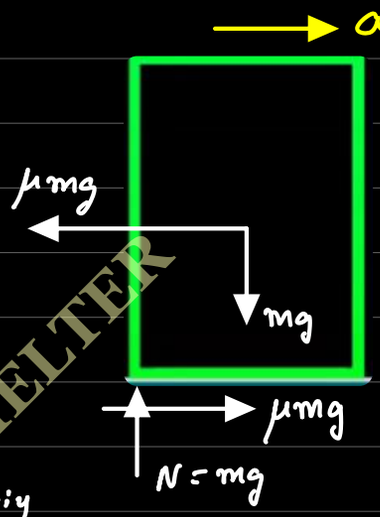
We see that slipping happens at smaller acceleration.

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Thus the Almirah will slip before toppling.

EXAMPLE: Suppose in the previous problem, Almirah begins to slip but the truck still continues to increase its acceleration, then at what acceleration will the Almirah also topple while slipping?

Solution: Acceleration of the Almirah from ground frame while slipping is $\mu g \rightarrow$ (Assuming truck is accelerating forward)



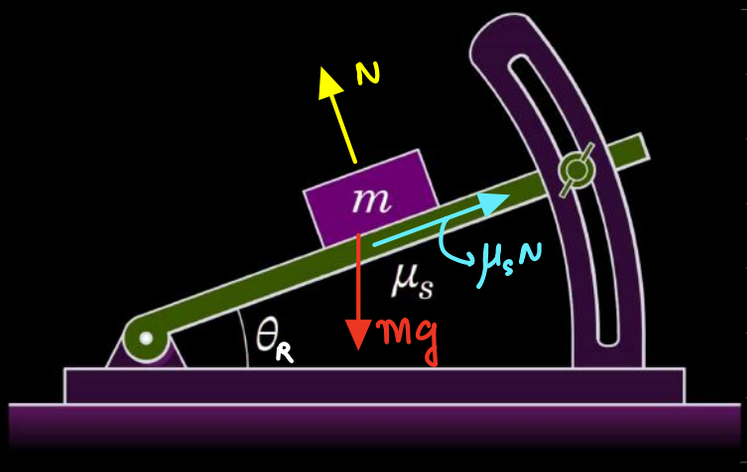
Now we draw the FBD of the Almirah from its own frame (see figure). For toppling, entire reaction will shift to the rear edge. Thus balancing torques:

$$\mu mg \times \frac{h}{2} = mg \times w/2 \quad \Rightarrow \quad \mu = w/h = 2/3 \text{ but } \mu = 1/2$$

Thus toppling will not happen.

→ Note carefully that here toppling is not possible no matter what the acceleration of the truck.

EXAMPLE: A uniform block of height h and width w is placed on an incline with gradually increasing inclination. If $w/h = 2$ and coefficient of friction is $\mu = 1/5$. Will the block first topple or slip?



Solution: For slipping $\theta = \theta_R$, thus

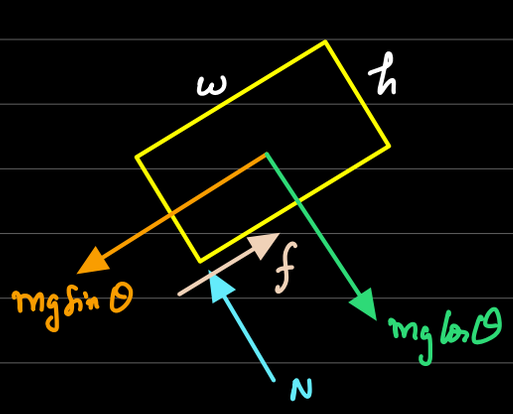
$\tan\theta_s = \mu = 1/5$. (s: slipping).

Now for toppling assuming no slipping, Torque of $mg\sin\theta$ about the bottom front edge must exceed that of $mg\cos\theta$.

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Thus: $mg\sin\theta \times \frac{h}{2} = mg\cos\theta \times \frac{w}{2}$

$\Rightarrow \quad \tan\theta_t = w/h = 2 \quad (t: \text{toppling})$

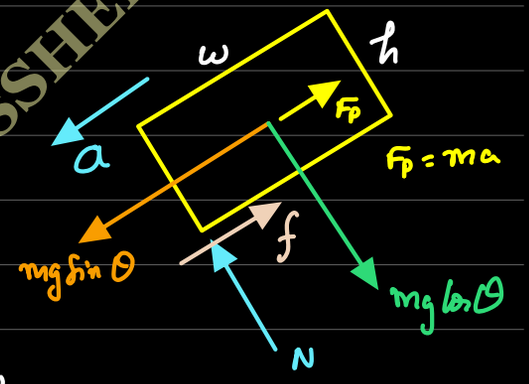


Thus $\tan\theta_s < \tan\theta_t$ therefore first slipping will occur, as the inclination is gradually increased.

EXAMPLE: Suppose in the previous situation, you continue to increase the inclination even after slipping has started, will there be toppling at any angle?

Solution: If slipping has already started, $a = g\sin\theta - \mu g\cos\theta$. Now for toppling, combined torque of $mg\sin\theta$ and pseudo force about front edge must be equal to torque of $mg\cos\theta$, thus:

$$[mg\sin\theta - m(g\sin\theta - \mu g\cos\theta)]\frac{h}{2} = mg\cos\theta \times \frac{w}{2}$$



$\Rightarrow \quad \mu mg\cos\theta\, h = mg\cos\theta\, w \Rightarrow \mu = \frac{w}{h}$ or $\frac{1}{s} = 2$ which is impossible.

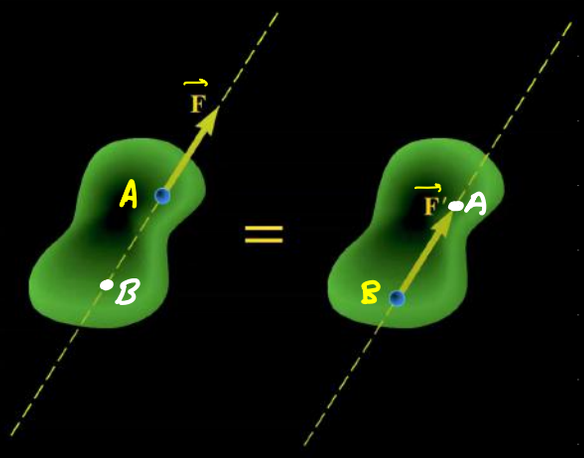
EQUIVALENT FORCE SYSTEMS (EQUIPOLLENT SYSTEMS):

Two or more force / couple systems are said to be equipollent or equivalent if for any rigid body they produce same net torque about every point and create same net acceleration of CM (or net resultant force is same). Visualisation in terms of equivalent force systems sometimes helps in simplifying the problems and gives some insights in to dynamical behaviours of systems. In the upcoming subsections we will see some equivalent transformations of force systems.

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PRINCIPLE OF TRANSMISSIBILITY:

A force acting on any rigid body can always be translated along its line of action to a different point and its torque about any arbitrary point will still remain the same. This can be easily seen as follows. Let $\vec{r}_A$ be position vector of point A in the first figure wrt some arbitrary point O, then its torque is given as, $\vec{\tau}_A = \vec{r}_A \times \vec{F}$ in



Two forces F and F′ are equivalent if they have the same magnitude and direction and the same line of action, even if they act at different points.

second case when the same force is translated to point B along same line, the torque becomes:

$$\vec{\tau}_B = \vec{r}_B \times \vec{F} = (\vec{r}_A + \vec{AB}) \times \vec{F} = \vec{r}_A \times \vec{F} + \vec{AB} \times \vec{F}$$

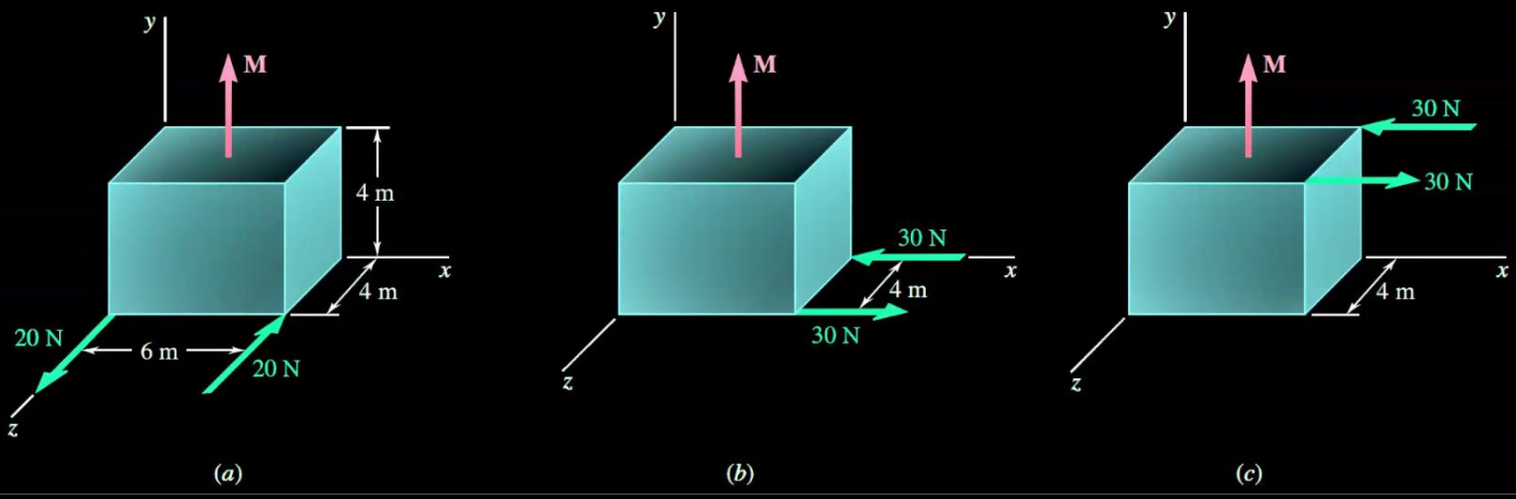
But since $\vec{F}$ is along $\vec{AB}$, the second term becomes zero. Thus:

$$\vec{\tau}_A = \vec{\tau}_B \quad \text{QED.}$$

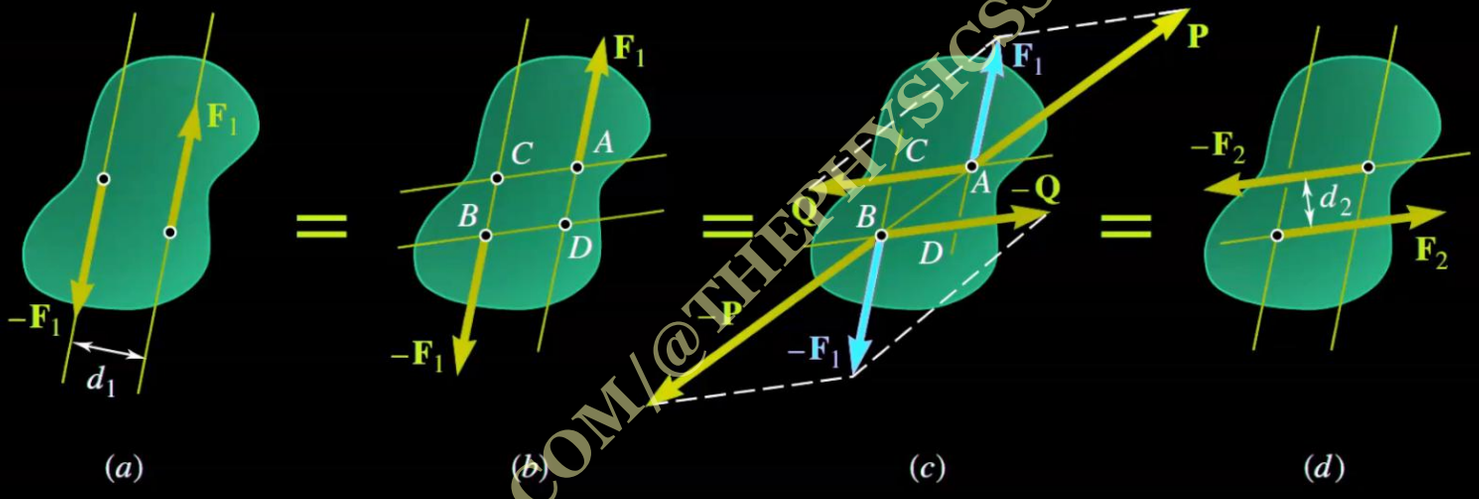
→ Note that this equivalence holds only for rigid bodies for the purpose of calculation of bulk motion. Clearly pulling somebody from his hairs upwards or pushing somebody upward by putting them on a lift floor are very different experiences as far as stresses in the tissues are concerned even if they produce same accelerations. (Savage! ☺).

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EQUIVALENT COUPLES:



Recall that torque of a couple is same about every point in the universe thus so long as the direction and magnitude of couple are maintained, the same couple can be generated by any two equal and opposite forces in a plane parallel to the plane containing forces of the original couple. Their magnitudes and separation can be adjusted so that net magnitude of couple remains the same. Thus all the three figures (a), (b) and (c) above represent equivalent couples from rigid body mechanics point of view. (You can see that in each case the plane of each pair is // to the plane of other pairs and product of forces and perpendicular distance between their lines of action are conserved.

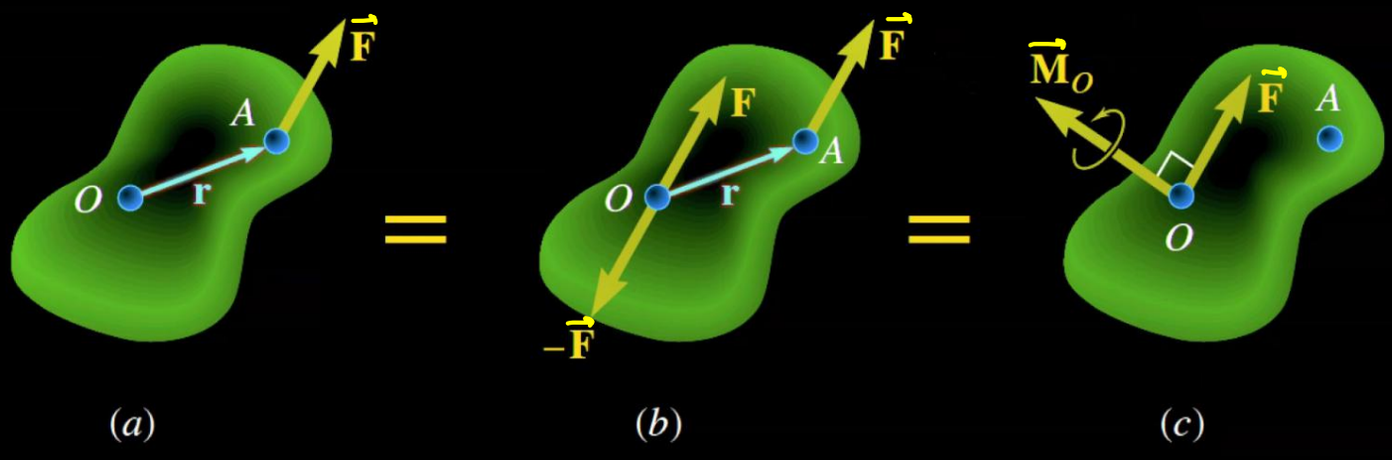


Four steps in transforming one couple to another couple in the same plane by using simple operations. (*a*) Starting couple; (*b*) label points of intersection of lines of action of the two couples; (*c*) resolve forces from first couple into components; (*d*) final couple.

A couple due to forces acting along two parallel lines can be transformed to a couple due to forces along another pair of lines coplanar with original parallel lines as per scheme shown in the above figure. Once we have the equivalent couple, now we can translate the couple anywhere in any plane parallel to itself. (Forces can always be translated along their lines of action).

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Resolving A Force In To Force And Couple At Arbitrary Point:



Replacing a force with a force and a couple. (*a*) Initial force **F** acting at point *A*; (*b*) attaching equal and opposite forces at *O*; (*c*) force **F** acting at point *O* and a couple.

If a force is translated parallel to its line of action to a different point, a couple has to be introduced to account for shift in the line of action as can be seen from the above schematic figure. It can be clearly seen that an additional couple of $\vec{r} \times \vec{F}$ must be introduced to keep the system equipollent.

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Example: Irodov 1.234:

A thin uniform rod AB of mass $m = 1.0$ kg moves translationally with acceleration $w = 2.0\ \text{ms}^{-2}$ due to two antiparallel forces $\mathbf{F}_1$ and $\mathbf{F}_2$ (Fig. 1.52). The distance between the points at which these forces are applied is equal to $a = 20$ cm. Besides, it is known that $F_2 = 5.0$ N. find the length of the rod.

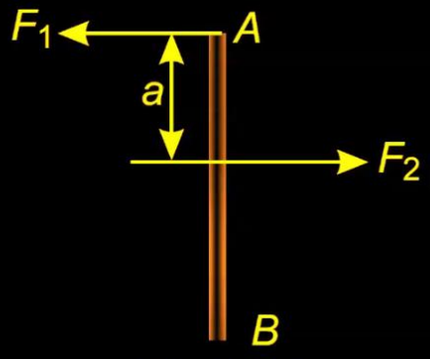
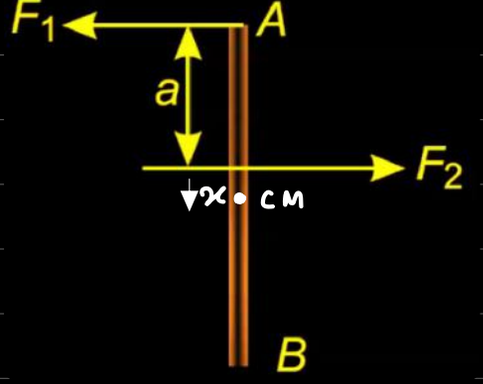


Fig. 1.52

Solution: As rod accelerates, a pseudo force will act it centre of mass from rest frame of the rod. Thus torque of real forces about this point must be zero (∵ torque of pseudo force is anyway zero). Now let CM be locate x distance below line of action of F_2. Thus by torque balance,

$F_2 x = F_1 (a + x)$ ① also $F_2 - F_1 = mw$ ②

Solving ① & ② we get,



Solving we set $F_2 = F_1 - mw$ and $x = \frac{a}{mw}(F_2 - mw)$.

Now length of rod : $l = 2(a+x) = 2aF_2/mw$ Ans

On putting values, $l = 1\,m$ Ans.

→ Note that even if the rod were non uniform, we could still locate the CM of the rod (x can still be solved).

EXAMPLE: IRODOV 1.235:

A force $\vec{\mathbf{F}} = A\mathbf{i} + B\mathbf{j}$ is applied to a point whose radius vector relative to the origin of coordinates O is equal to $\vec{\mathbf{r}} = a\mathbf{i} + b\mathbf{j}$, where a, b, A, B are constants, and $\mathbf{i}$, $\mathbf{j}$ are the unit vectors of the x and y axes. Find the moment $\mathbf{N}$ and the arm l of the force $\mathbf{F}$ relative to the point O.

Solution: Here arm means ⊥ distance of line of action of force from the point. Now recall that moment of a force is simply $\tau = r_\perp F$ thus $r_\perp = \tau/F$. Now,

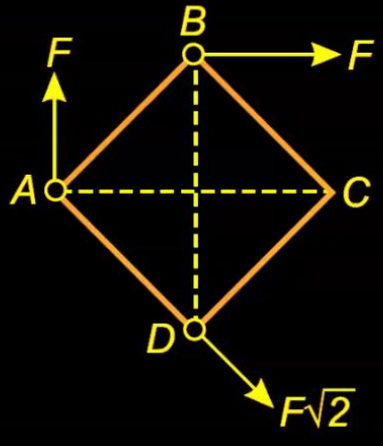
$$\vec{\tau} = \vec{r} \times \vec{F} = (A\hat{i} + B\hat{j}) \times (a\hat{i} + b\hat{j}) = (aB - Ab)\hat{k} \text{ Ans}$$

Now:

$$r_\perp = \tau/F = (aB - Ab)/\sqrt{A^2 + B^2} \quad \text{Ans.}$$

EXAMPLE: IRODOV 1.237:

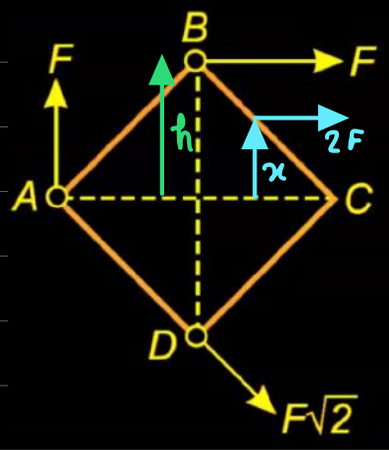
Three forces are applied to a square plate as shown in Fig. 1.53. Find the modulus, direction, and the point of application of the resultant force, if this point is taken on the side BC.



Solution: Here we wish to replace all the forces by a single force somewhere on BC such that torque of this force is same as that of the original force system about any point while maintaining same resultant force.

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Here the resultant force is clearly 2F to the right so we need to apply 2F rightwards somewhere on BC. Let this force act at a height x above the line AC. Now conserving torque about point A, $Fh = 2Fx \Rightarrow x = h/2$



Thus the equivalent force must be applied at mid point of BC with a magnitude 2F to produce the same dynamical effect.

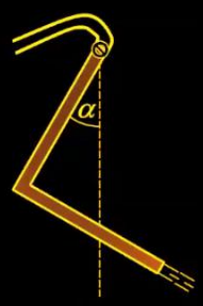
→ Here A was a good choice for torque equivalence because two forces passed through it thus reducing our calculation.

→ Since equivalent systems produce same torque about every point, this gives us freedom in choosing the point for purpose of calculating the effective point of application of force.

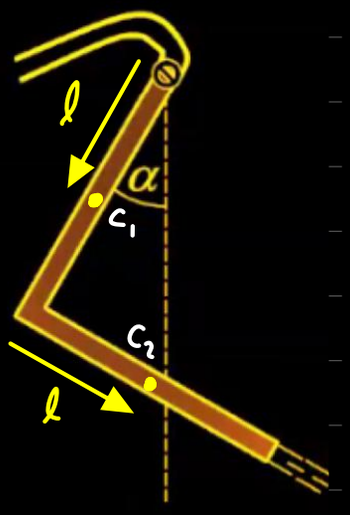
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EXAMPLE: PATHFINDER: BUILD 24:

A metal pipe bent at right angle at its midpoint is pivoted to a fixed support at one of its ends so that it can rotate freely in its plane about a horizontal axis through the pivoted end. A flexible hose is connected at the pivoted end. When water is fed at a particular rate into the pipe, the upper arm of the pipe makes an angle α with the vertical as shown in the figure. If the rate of feeding water is gradually made η times, find an expression for the angle β, which the upper arm of the pipe will make with the vertical.



Solution: Here gravitational torque about the pivot is responsible rate of creation of angular momentum. Let v be the speed of water, ρ its density and S cross section of the pipe then in time dt, angular momentum created is $(\rho S v dt)\, v \times \underbrace{2\ell}_{r_\perp}$ ($\uparrow dm$)



i.e $\frac{dL}{dt} = 2\rho S \ell v^2$, this must be equal to

gravitational torque, $\tau_G = (\rho S \times 2\ell) g\, \ell \sin\alpha + (\rho S \times 2\ell) g \left[2\ell \sin\alpha - \ell \sin\alpha\right)$

Thus: $2\rho S \ell v^2 = 2\rho S \ell^2 g\, [3\sin\alpha - \cos\alpha]$

$\Rightarrow \quad 3\sin\alpha - \cos\alpha = v^2/lg = \sqrt{10}\sin(\alpha - \tan^{-1}1/3)$ ①

Similarly when flow rate is made η times and angle becomes β:

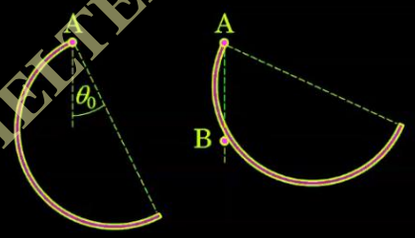
$\sqrt{10}\sin[\beta - \tan^{-1}1/3) = (\eta v)^2/lg$ ② thus from ① & ②

$$\frac{\sin(\beta - \tan^{-1}1/3)}{\sin(\alpha - \tan^{-1}1/3)} = \eta^2 \Rightarrow \beta = \tan^{-1}(1/3) + \sin^{-1}\left(\eta^2 \sin|\alpha - \tan^{-1}1/3\right).$$

Ans.

EXAMPLE: PATHFINDER: CHECK-6:

One end of a half ring of mass m is hinged to nail A. When the ring is set free, its diameter makes an angle θ_0 with the vertical as shown in the first figure. Another nail is now driven into the wall, vertically below the former nail at a distance that is equal to radius of the ring and the ring is left resting on the lower nail as shown in the second figure. Assuming friction to be absent everywhere and acceleration due to gravity to be g, find horizontal and vertical components of the reaction force of the upper nail on the ring in the second case.



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Solution: Torque balance about A:

$mg\sin(60° - \theta_0) \times R\cos\theta_0 = N\sin 60° \times R$ ①

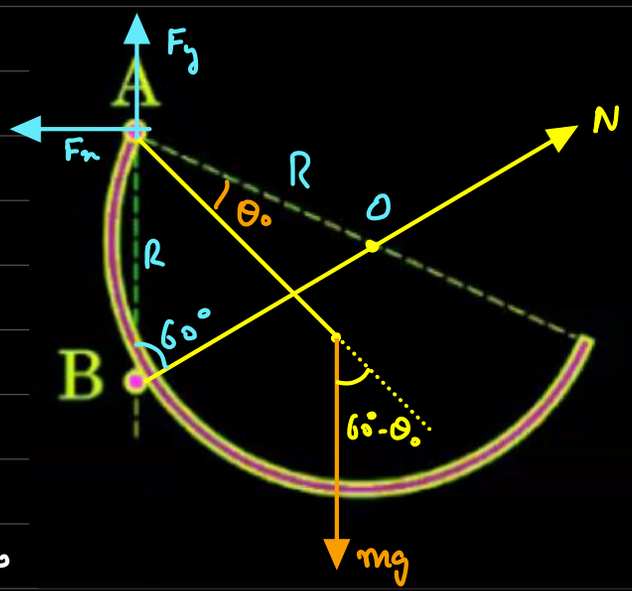
Horizontal force balance:

$F_x = N\sin 60°$ ②

From ① & ②:

$F_x = mg\sin(60° - \theta_0)\cos\theta_0$

$= mg\frac{\sqrt{3}}{2}\left[1 - \frac{1}{\sqrt{3}}\tan\theta\right]$ Ans.

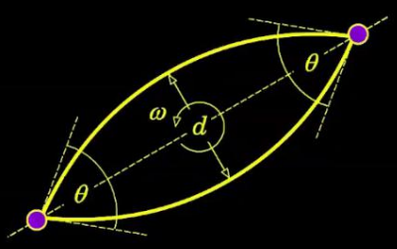


Now $F_y = mg - N\cos 60°$

$= mg - mg\sin(60° - \theta_0)\cos\theta_0\cot 60°$

$= \frac{mg}{2}\left[1 + \frac{\tan\theta_0}{\sqrt{3}}\right]$ Ans.

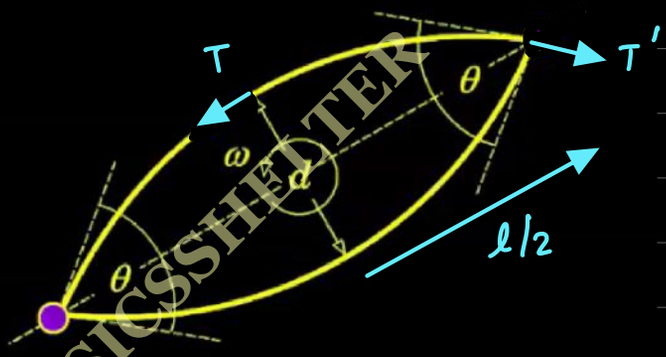
Example: Pathfinder: Check-1s:



In free space, an assembly of two identical particles each of mass m connected by two identical uniform ropes is rotating at a constant angular velocity ω about an axis passing through the centre of the line segment joining the particles. The ropes fan out due to inertia and assume a curved shape as shown in the figure. If distance between the midpoints of the ropes is d, tensile force at the midpoint of a rope is T and the included angle between the ropes at the particles is θ, find the distance l between the particles.

Solution: Consider FBD of the portion of the rope as shown. Let the tension be T and T'. Centrifugal forces produce no torque about centre of rotation. Thus balancing torques about centre:

$$T\frac{d}{2} = T'\sin\left(\frac{\theta}{2}\right) \times l/2 \quad \text{①}$$



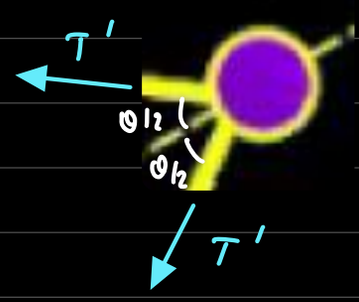
Now considering FBD of the particles:

$$2T'\cos(\theta/2) = m\omega^2(l/2) \quad \text{②}$$

From ① & ②,

$$\frac{Td}{m\omega^2 l} = \frac{\tan(\theta/2)\,l}{4}$$

$$\Rightarrow l = \frac{2}{\omega}\sqrt{\frac{Td}{m\tan(\theta/2)}} \quad \text{Ans.}$$



Example: Pathfinder: Check 22:

Very often, you have seen an insect dragging an object much larger and heavier than it. The figure is a top view of an ant dragging a straw that is much larger and heavier than the ant. The ant pulls the straw at one end rotating it slightly then it does the same action at the other end and so on.



(a) How does this technique work?

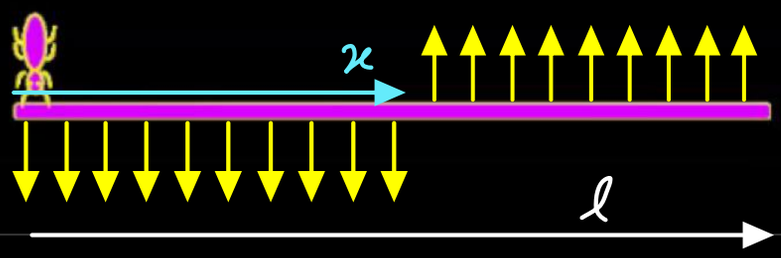
(b) An ant is dragging a straw of mass m on a horizontal floor by this technique. Coefficient of friction between the straw and the floor is μ. How much force F does it have to apply?

Solution: When the straw is at the verge of slipping by torque balance considerations it can be clearly seen that frictional force on the straw cannot be in same direction everywhere otherwise

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its torque about the ant will become non-zero. Let the friction change its direction at a distance x and let l be length of the straw.

Now balancing torques about the ant:



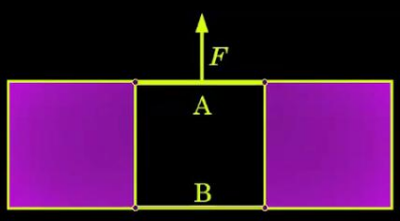
$$\mu mg\frac{x}{l}\times\frac{x}{2}=\frac{\mu mg(l-x)}{l}\times\left(\frac{l+x}{2}\right)$$

$\Rightarrow x = l/\sqrt{2}$, Now net frictional force: $\mu mg\frac{x}{l} - \mu mg\frac{(l-x)}{l}$

$$= \frac{\mu mg}{l}[2x-l] = \mu mg(\sqrt{2}-1) \text{ Ans.}$$

→ Note that frictional field was piecewise uniform so we could assume friction to be at the respective mid points for torque calculations.

Example: Pathfinder: Build 26:



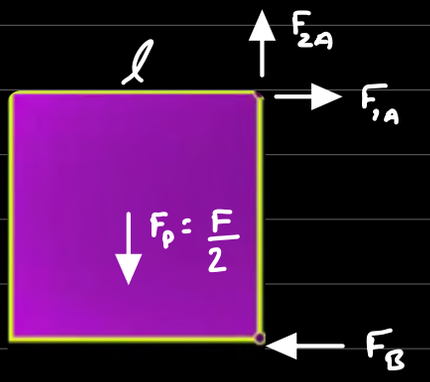
Two identical square laminae connected by two rods A and B are placed on a frictionless horizontal floor with their flat faces on the floor. The rods are free to rotate at the joints where they are connected with the laminae. The rod A is pulled horizontally at its midpoint perpendicular to the rod by a constant force F as shown in the figure. Find moduli of the reaction forces F_A and F_B between one of the laminae and the rods A and B respectively. Consider the following cases.

(a) Mass of each rod is negligible as compared to that of a lamina.
(b) Mass of each rod is equal to that of a lamina.

Solution: Here one of the key ideas is that whenever a massless rod is loaded only at the pin joints at is ends, the force exerted by the rod is along the rod. (For example rod B in part a).

(a) Now consider FBD of left lamina in its rest frame. Balancing torques about upper hinge:

$$\frac{F}{2}\times\frac{l}{2} = F_B\times l \Rightarrow F_B = F/4 \text{ Ans.}$$



By horizontal force balance, $F_{1A} = F_B = F/4$

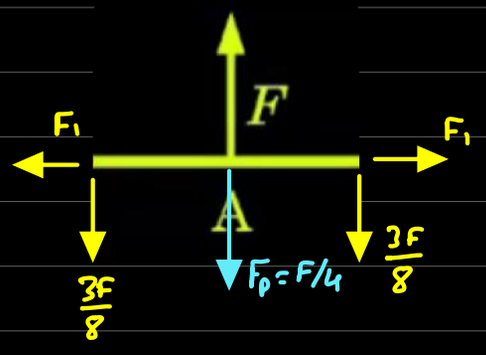
By vertical force balance $F_{2A} = \frac{F}{2}$ thus,

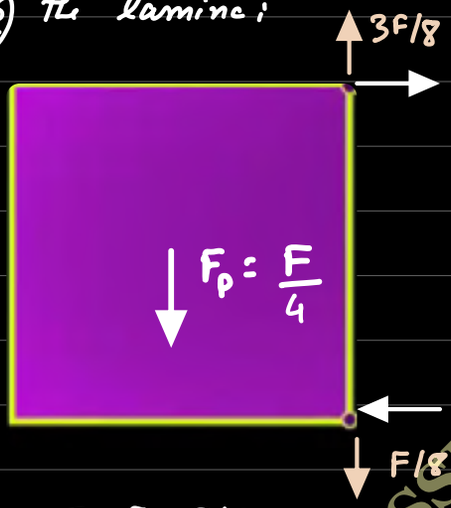
$$F_A = \sqrt{(F/4)^2+(F/2)^2} = \frac{\sqrt{5}}{4}F \text{ Ans.}$$

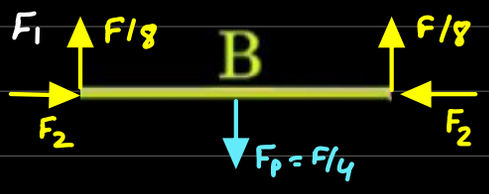
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(b) Now when rods are massive, $a = F/4m$.

FBD of rods from their rest frame are shown. (Vertical force balance gives the vertical forces at the ends). Now Consider FBD of the lamina:







Now balancing torque about lower hinge from rest frame of lamina,

$F_1 \times \ell = \frac{F}{4} \times \frac{\ell}{2}$

$\Rightarrow F_1 = F/8$

Also by horizontal force balance, $F_2 = F_1 = F/8$.

Thus hinge reactions: $F_A = \sqrt{(3F/8)^2 + (F/8)^2} = \frac{\sqrt{10}}{8} F$ Ans.

$F_B = \sqrt{(F/8)^2 + (F/8)^2} = \frac{\sqrt{2}}{8} F$ Ans.

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